CHAPTER 22
Carbonyl Alpha-Substitution Reactions



FIGURE 22.1 **Tear gas is a controversial riot-control device used by some military and police forces.** It is a simple chloroketone made by a carbonyl α-substitution reaction. (credit: modification of work "Tear Gas in CWB Hennessy Road View" by The Stand News/Wikimedia Commons, Public Domain)

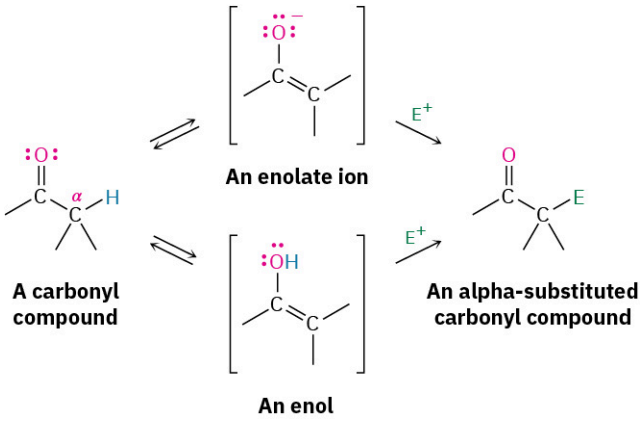
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WHY THIS CHAPTER? As with nucleophilic additions and nucleophilic acyl substitutions, many laboratory schemes, pharmaceutical syntheses, and biochemical pathways make frequent use of carbonyl α-substitution reactions. Their great value comes from the fact that they constitute one of the few general methods for forming carbon–carbon bonds, thereby making it possible to build larger molecules from smaller precursors. In this chapter, we'll see how and why these reactions occur.

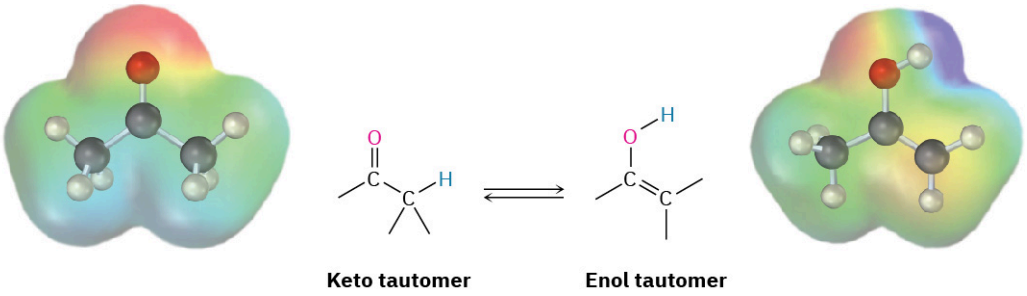
We said in the ***Preview of Carbonyl Chemistry*** that much of the chemistry of carbonyl compounds can be explained by just four fundamental reaction types: nucleophilic additions, nucleophilic acyl substitutions, α substitutions, and carbonyl condensations. Having studied the first two of these reactions in the past three chapters, let's now look in more detail at the third major carbonyl-group process—the **α-substitution reaction**.

Alpha-substitution reactions occur at the position next to the carbonyl group—the α *position*—and involve the substitution of an α hydrogen atom by an electrophile, E, through either an enol or enolate ion intermediate. Let's begin by learning more about these two species.



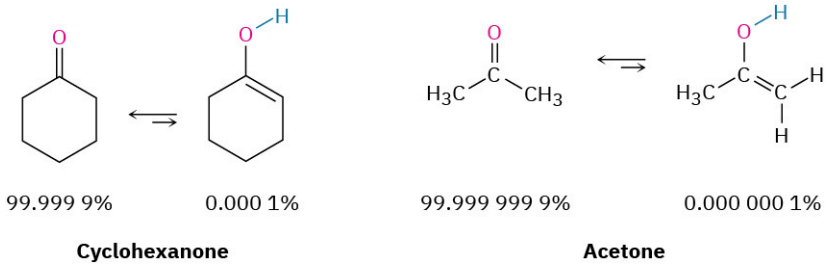
22.1 Keto–Enol Tautomerism

A carbonyl compound with a hydrogen atom on its α carbon is in equilibrium with its corresponding **enol** isomer (Section 9.4). This spontaneous interconversion between two isomers, usually with the change in position of a hydrogen, is called *tautomerism,* from the Greek *tauto,* meaning "the same," and *meros,* meaning "part." The individual keto and enol isomers are called **tautomers**.



Note the difference between tautomers and resonance forms. Tautomers are constitutional isomers—different compounds with different structures—while resonance forms are different representations of a single compound. Tautomers have their *atoms* arranged differently, while resonance forms differ only in the position of their π and nonbonding *electrons.*

Most monocarbonyl compounds exist almost entirely in their keto form at equilibrium, and it's usually difficult to isolate the pure enol. Cyclohexanone, for example, contains only about 0.0001% of its enol tautomer at room temperature. The percentage of enol tautomer is even lower for carboxylic acids, esters, and amides. The enol can only predominate when stabilized by conjugation or intramolecular hydrogen bonding. Thus, 2,4-pentanedione is about 76% enol tautomer. Although enols are scarce at equilibrium, they are nevertheless responsible for much of the chemistry of carbonyl compounds because they are so reactive.



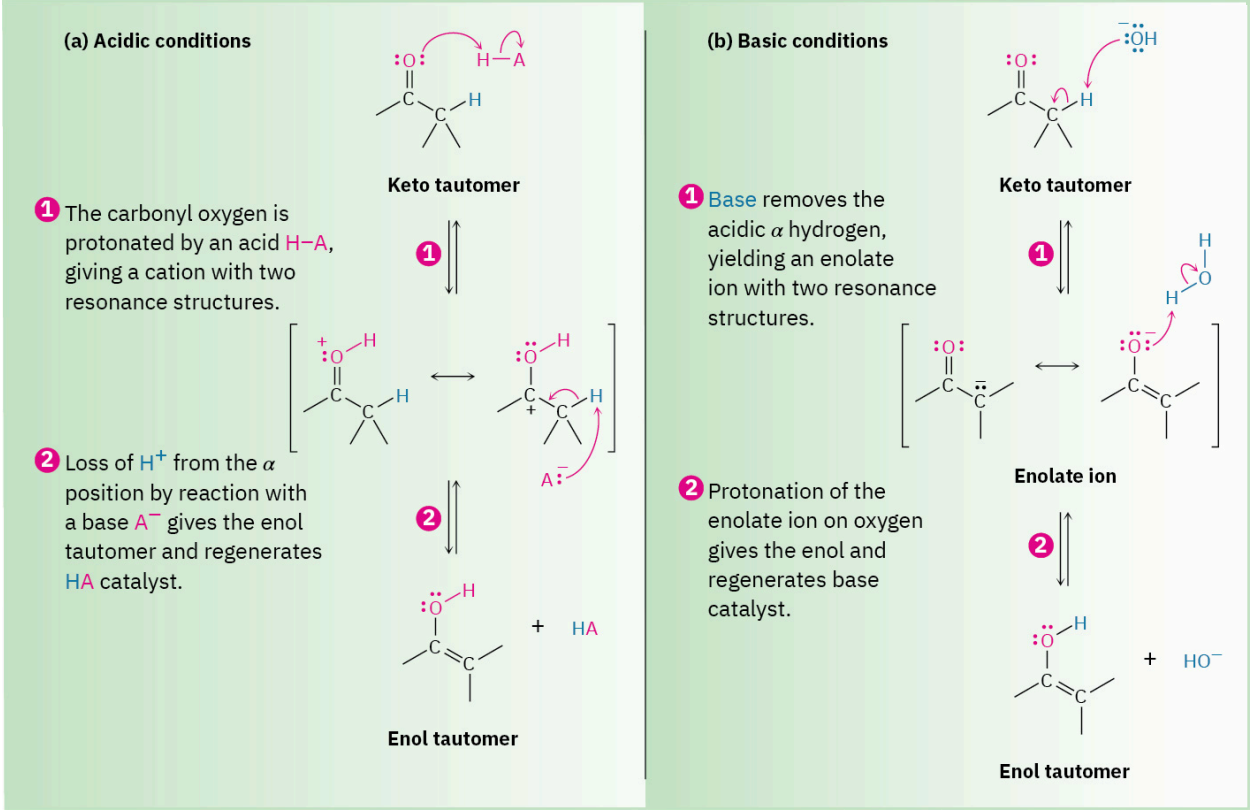
Keto–enol tautomerism of carbonyl compounds is catalyzed by both acids and bases. Acid catalysis occurs by protonation of the carbonyl oxygen atom to give an intermediate cation that loses H^+ from its α carbon to yield a neutral enol (FIGURE 22.2a). This proton loss from the cation intermediate is similar to what occurs during an E1 reaction when a carbocation loses H^+ to form an alkene (Section 11.10).

Base-catalyzed enol formation occurs because the presence of a carbonyl group makes the hydrogens on the α carbon weakly acidic. Thus, a carbonyl compound can act as an acid and donate one of its α hydrogens

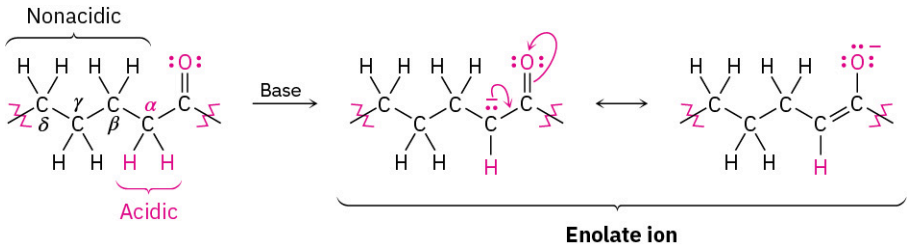
to a sufficiently strong base. The resultant resonance-stabilized anion, an **enolate ion**, is then protonated to yield a neutral compound. If protonation of the enolate ion takes place on the α carbon, the keto tautomer is regenerated and no net change occurs. If, however, protonation takes place on the oxygen atom, an enol tautomer is formed (FIGURE 22.2b).

FIGURE 22.2 MECHANISM

Mechanism of enol formation under both acid-catalyzed and base-catalyzed conditions. (a) Acid catalysis involves (1) initial protonation of the carbonyl oxygen followed by (2) removal of H^+ from the α position. **(b)** Base catalysis involves (1) initial deprotonation of the α position to give an enolate ion, followed by (2) reprotonation on oxygen.



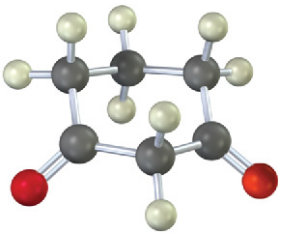
Note that only the hydrogens on the α position of carbonyl compounds are acidic. Hydrogens at β, γ, δ, and so on, aren't acidic and can't be removed by base because the resulting anions can't be resonance-stabilized by the carbonyl group.



PROBLEM 22-1 Draw structures for the enol tautomers of the following compounds:
(a) Cyclopentanone **(b)** Methyl thioacetate **(c)** Ethyl acetate **(d)** Propanal **(e)** Acetic acid **(f)** Phenylacetone

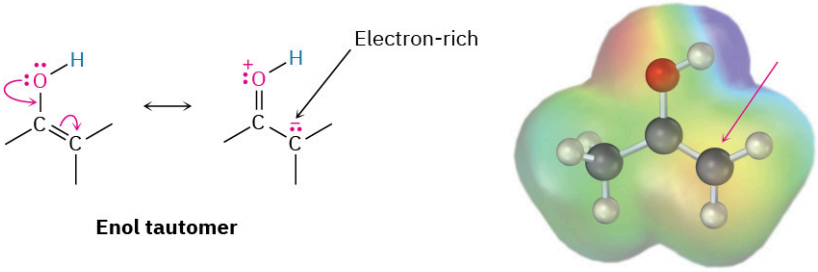
PROBLEM 22-2 How many acidic hydrogens does each of the molecules listed in Problem 22-1 have? Identify them.

PROBLEM 22-3 Draw structures for all monoenol forms of the following molecule. Which would you expect to be most stable? Explain.



22.2 Reactivity of Enols: α-Substitution Reactions

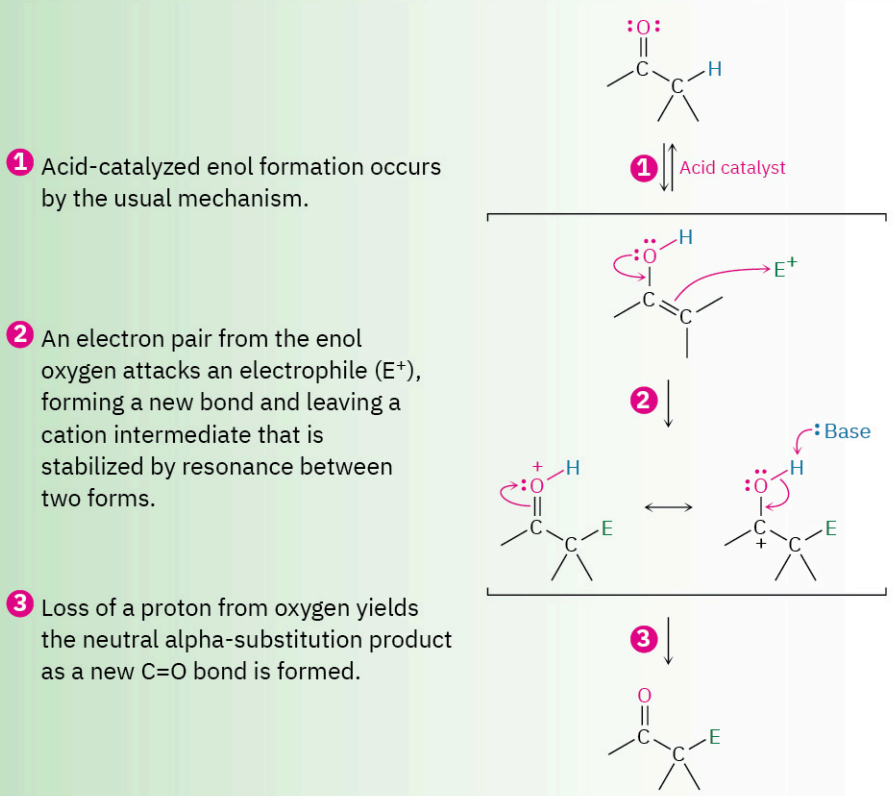
What kind of chemistry do enols have? Because their double bonds are electron-rich, enols behave as nucleophiles and react with electrophiles in much the same way that alkenes do. But because of resonance-electron donation of a lone pair of electrons on the neighboring oxygen, enols are more electron-rich and correspondingly more reactive than alkenes. Notice in the following electrostatic potential map of ethenol ($H_2C{=}CHOH$) how there is a substantial electron density (yellow-red) on the α carbon.



When an *alkene* reacts with an electrophile, E^+, initial addition gives an intermediate cation and subsequent reaction with a nucleophile, such as a halide ion, yields an addition product (Section 7.7). When an *enol* reacts with an electrophile, however, only the initial addition step is the same (FIGURE 22.3). Instead of reacting with a nucleophile to give an addition product, the intermediate cation loses the –OH proton to give an α-substituted carbonyl compound.

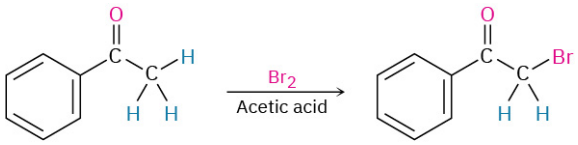
FIGURE 22.3 MECHANISM

General mechanism of a carbonyl α-substitution reaction. In step 3, the initially formed cation loses H^+ to regenerate a carbonyl compound.



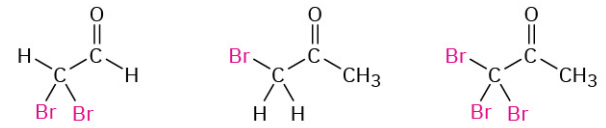
22.3 Alpha Halogenation of Aldehydes and Ketones

A particularly common α-substitution reaction in the laboratory is the halogenation of aldehydes and ketones at their α positions by reaction with Cl_2, Br_2, or I_2 in acidic solution. Bromine in acetic acid solvent is often used.



Acetophenone **α-Bromoacetophenone (72%)**

Remarkably, ketone halogenation also occurs in biological systems, particularly in marine alga, where dibromoacetaldehyde, bromoacetone, 1,1,1-tribromoacetone, and other related compounds have been found.

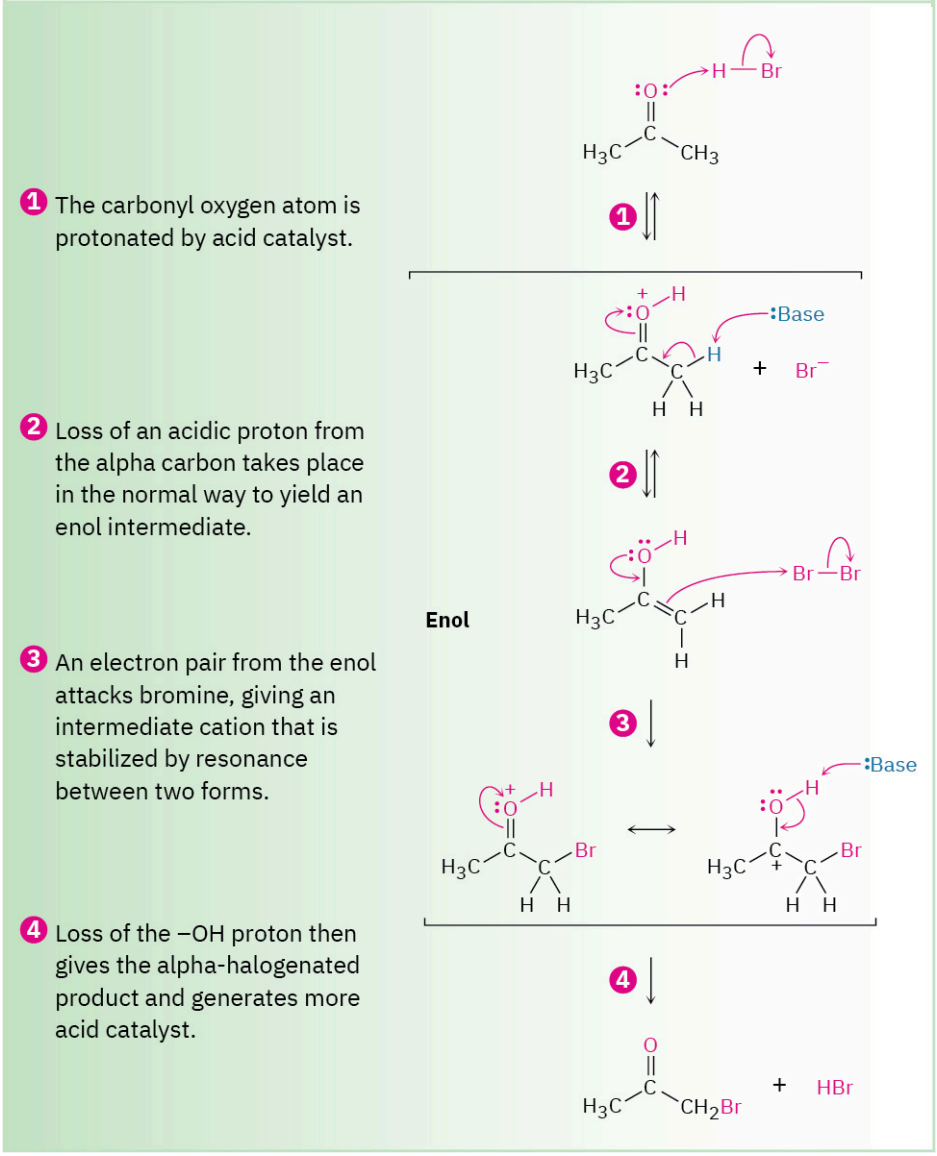


From the Hawaiian alga *Asparagopsis taxiformis*

This form of halogenation is a typical α-substitution reaction that proceeds by acid-catalyzed formation of an enol intermediate, as shown in FIGURE 22.4.

FIGURE 22.4 MECHANISM

Mechanism of the acid-catalyzed bromination of acetone.

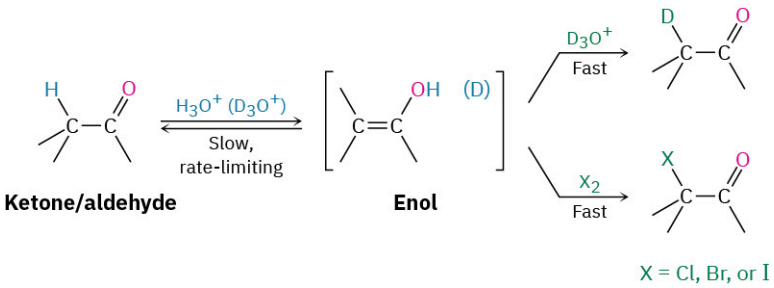


Evidence for the mechanism shown in **FIGURE 22.4** includes the observation that acid-catalyzed halogenations show second-order kinetics and follow the rate law

$$\text{Reaction rate} = k[\text{Ketone}][\text{H}^+]$$

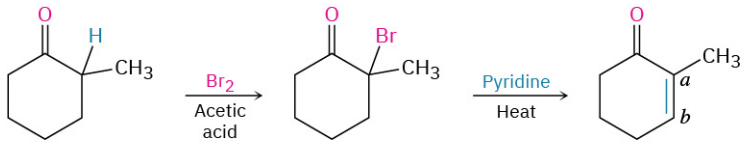
In other words, the rate of halogenation depends only on the concentrations of ketone and acid and is independent of halogen concentration. Since halogen is not involved in the rate-limiting step, chlorination, bromination, and iodination of a given substrate all occur at the same rate.

Furthermore, if an aldehyde or ketone is treated with D_3O^+, the acidic α hydrogens are replaced by deuterium. For a given ketone, the rate of deuterium exchange is identical to the rate of halogenation, implying that a common intermediate—presumably the enol—is involved in both processes.



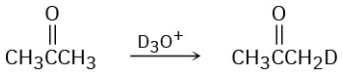
Ketone/aldehyde **Enol**

α-Bromo ketones are useful in the laboratory because they can be dehydrobrominated by base treatment to yield α,β-unsaturated ketones. For example, 2-methylcyclohexanone gives 2-bromo-2-methylcyclohexanone on halogenation, and the α-bromo ketone gives 2-methyl-2-cyclohexenone when heated in pyridine. The reaction takes place by an E2 elimination pathway (Section 11.8) and is a good method for introducing a C=C bond into a molecule. Note that bromination of 2-methylcyclohexanone occurs primarily on the more highly substituted α position because the more highly substituted enol is favored over the less highly substituted one (Section 7.6).



2-Methylcyclo-hexanone **2-Bromo-2-methyl-cyclohexanone** **2-Methyl-2-cyclo-hexenone (63%)**

PROBLEM 22-4 Write the complete mechanism for the deuteration of acetone on treatment with D_3O^+.



PROBLEM 22-5 Show how you might prepare 1-penten-3-one from 3-pentanone.

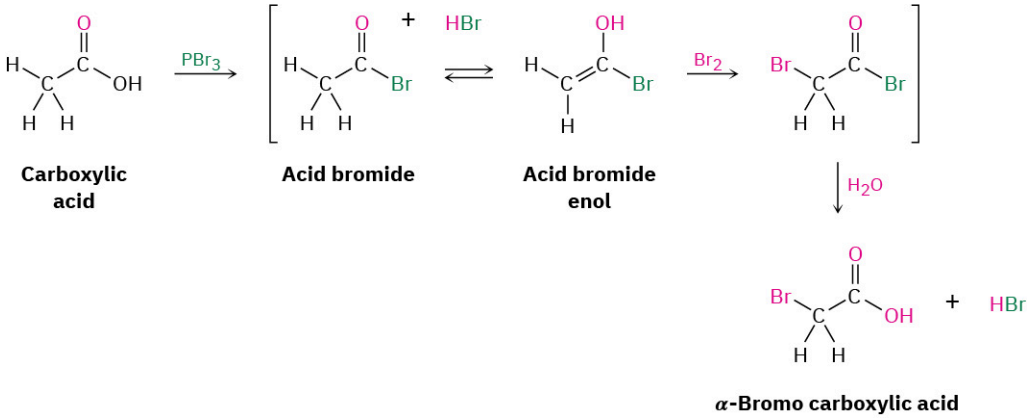
22.4 Alpha Bromination of Carboxylic Acids

The α bromination of carbonyl compounds by Br_2 in acetic acid is limited to aldehydes and ketones because acids, esters, and amides don't enolize to a sufficient extent. Carboxylic acids, however, can be α brominated by a mixture of Br_2 and PBr_3 in the *Hell–Volhard–Zelinskii (HVZ) reaction.*

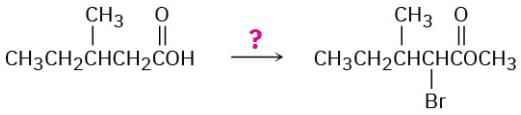


Heptanoic acid **2-Bromoheptanoic acid (90%)**

The Hell–Volhard–Zelinskii reaction is a bit more complex than it looks and actually involves α substitution of an acid bromide enol rather than a carboxylic acid enol. The process begins by reaction of the carboxylic acid with PBr_3 to form an acid bromide plus HBr (Section 21.4). The HBr then catalyzes enolization of the acid bromide, and the resultant enol reacts with Br_2 in an α-substitution reaction to give an α-bromo acid bromide. Addition of water hydrolyzes the acid bromide in a nucleophilic acyl substitution reaction and yields the α-bromo carboxylic acid product.



PROBLEM 22-6 If methanol rather than water is added at the end of a Hell–Volhard–Zelinskii reaction, an ester rather than an acid is produced. Show how you would carry out the following transformation, and propose a mechanism for the ester-forming step.



22.5 Acidity of Alpha Hydrogen Atoms: Enolate Ion Formation

As noted in **Section 22.1**, a hydrogen on the α position of a carbonyl compound is weakly acidic and can be removed by a strong base to yield an enolate ion. In comparing acetone ($pK_a = 19.3$) with ethane ($pK_a \approx 60$), for instance, the presence of a neighboring carbonyl group increases the acidity of the ketone over the alkane by a factor of 10^{40}.



Acetone ($pK_a = 19.3$) **Ethane ($pK_a \approx 60$)**

Proton abstraction from a carbonyl compound occurs when the α C–H bond is oriented roughly parallel to the p orbitals of the carbonyl group. The α carbon atom of the enolate ion is sp^2-hybridized and has a p orbital that overlaps the neighboring carbonyl p orbitals. Thus, the negative charge is shared by the electronegative oxygen atom, and the enolate ion is stabilized by resonance (**FIGURE 22.5**).

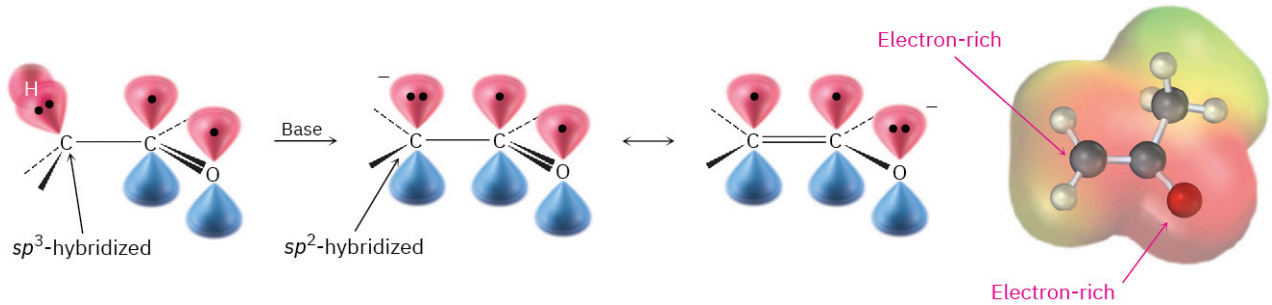
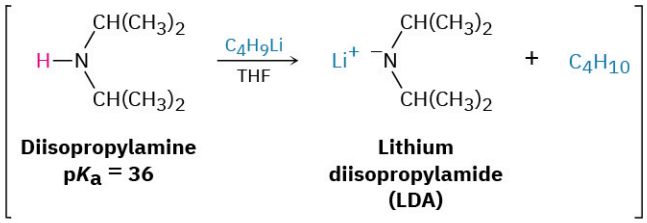


FIGURE 22.5 Mechanism of enolate-ion formation by abstraction of an α proton from a carbonyl compound. The enolate ion is stabilized by resonance, and the negative charge (red) is shared by the oxygen and the α carbon atom, as indicated in the electrostatic potential map.

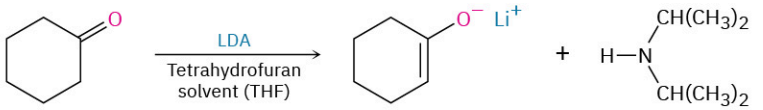
Because carbonyl compounds are only weakly acidic, a strong base is needed for the formation of an enolate-ion. If an alkoxide ion, such as sodium ethoxide, is used as base, deprotonation of acetone only takes place to the extent of about 0.1% because acetone is a weaker acid than ethanol ($pK_a = 16$). If, however, a more powerful base is used, then a carbonyl compound is completely converted into its enolate ion.

In practice, the strong base lithium diisopropylamide [$LiN(i\text{-}C_3H_7)_2$; abbreviated LDA] is commonly used for making enolate ions. As the lithium salt of the weak acid diisopropylamine, pK_a = 36, LDA can readily deprotonate most carbonyl compounds. It is easily prepared by reaction of butyllithium with diisopropylamine and is soluble in organic solvents because of its two alkyl groups.



Diisopropylamine
pK_a = 36

Lithium diisopropylamide (LDA)



Cyclohexanone

Cyclohexanone enolate ion

Diisopropylamine

Many types of carbonyl compounds, including aldehydes, ketones, esters, thioesters, acids, and amides, can be converted into enolate ions by reaction with LDA. **TABLE 22.1** lists the approximate pK_a values of different types of carbonyl compounds and shows how these values compare to other acidic substances we've seen. Note that nitriles, too, are acidic and can be converted into enolate-like anions.

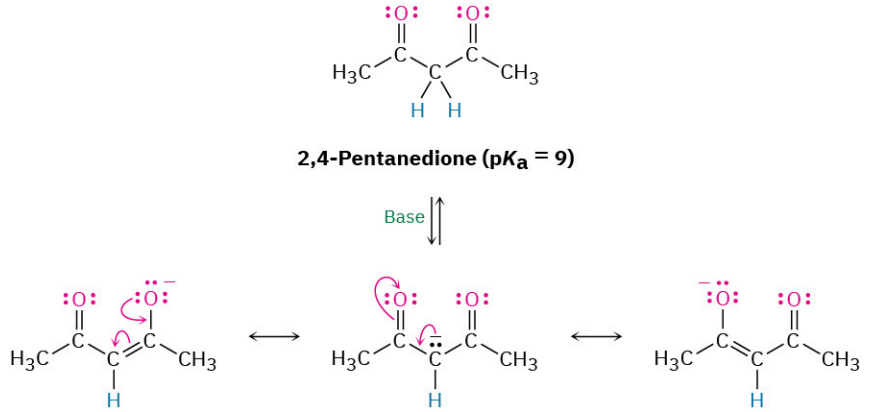
TABLE 22.1 Acidity Constants for Some Organic Compounds

Functional group	Example	pK_a
Carboxylic acid	CH_3COH	5
1,3-Diketone	$CH_3CCH_2CCH_3$	9
3-Keto ester	$CH_3CCH_2COCH_3$	11
1,3-Diester	$CH_3OCCH_2COCH_3$	13
Alcohol	CH_3OH	16
Acid chloride	CH_3CCl	16
Aldehyde	CH_3CH	17
Ketone	CH_3CCH_3	19

TABLE 22.1 Acidity Constants for Some Organic Compounds

Functional group	Example	pK_a
Thioester	CH_3CSCH_3	21
Ester	CH_3COCH_3	25
Nitrile	$CH_3C{\equiv}N$	25
N,*N*-Dialkylamide	$CH_3CN(CH_3)_2$	30
Dialkylamine	$HN(i\text{-}C_3H_7)_2$	36

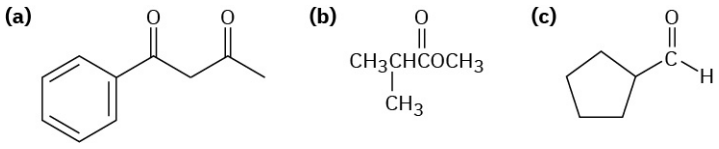
When a hydrogen atom is flanked by two carbonyl groups, its acidity is enhanced further. TABLE 22.1 thus shows that 1,3-diketones (β-diketones), 3-oxo esters (β-keto esters), and 1,3-diesters (malonic esters) are more acidic than water. The enolate ions derived from these β-dicarbonyl compounds are stabilized by sharing the negative charge of the two neighboring carbonyl oxygens. The enolate ion of 2,4-pentanedione, for instance, has three resonance forms. Similar resonance forms can be drawn for other doubly stabilized enolate ions.



WORKED EXAMPLE 22.1

Identifying the Acidic Hydrogens in a Compound

Identify the most acidic hydrogens in each of the following compounds, and rank the compounds in order of increasing acidity:

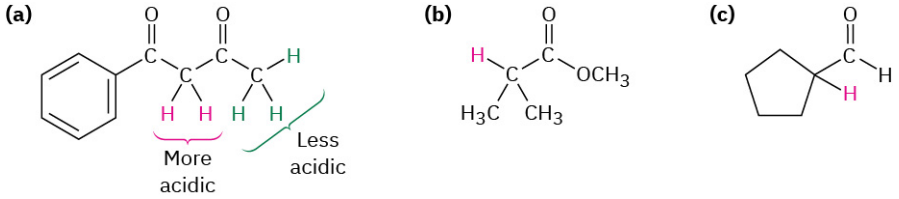


Strategy

Hydrogens on carbon next to a carbonyl group are acidic. In general, a β-dicarbonyl compound is most acidic, a ketone or aldehyde is next most acidic, and a carboxylic acid derivative is least acidic. Remember that alcohols, phenols, and carboxylic acids are also acidic because of their –OH hydrogens.

Solution

The acidity order is (**a**) > (**c**) > (**b**). Acidic hydrogens are shown in red.



PROBLEM 22-7 Identify the most acidic hydrogens in each of the following molecules:
(a) CH_3CH_2CHO **(b)** $(CH_3)_3C–COCH_3$ **(c)** CH_3CO_2H **(d)** Benzamide **(e)** $CH_3CH_2CH_2CN$ **(f)** $CH_3CON(CH_3)_2$

PROBLEM 22-8 Draw a resonance structure of the acetonitrile anion, $^-:CH_2C{\equiv}N$, and account for the acidity of nitriles.

22.6 Reactivity of Enolate Ions

Enolate ions are more useful than enols for two reasons. First, pure enols can't normally be isolated but are instead generated only as short-lived intermediates in low concentration. By contrast, stable solutions of pure enolate ions are easily prepared from most carbonyl compounds by reaction with a strong base. Second, enolate ions are more reactive than enols and undergo many reactions that enols don't. Whereas enols are neutral, enolate ions are negatively charged, making them much better nucleophiles.

Because they are resonance hybrids of two nonequivalent forms, enolate ions can be looked at either as vinylic alkoxides ($C{=}C–O^-$) or as α-keto carbanions ($^-C–C{=}O$). Thus, enolate ions can react with electrophiles either on oxygen or on carbon. Reaction on oxygen yields an enol derivative, while reaction on carbon yields an α-substituted carbonyl compound (FIGURE 22.6). Both kinds of reactivity are known, but reaction on carbon is more common.

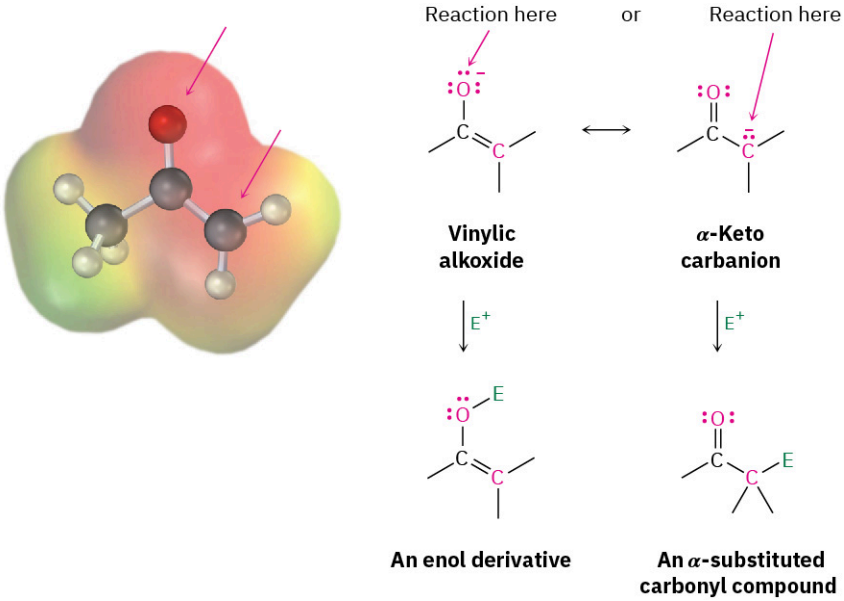
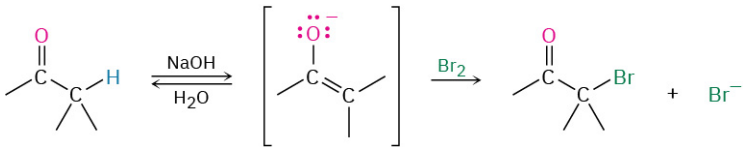


FIGURE 22.6 The electrostatic potential map of acetone enolate ion shows how the negative charge is delocalized over both the oxygen and the α carbon. As a result, two modes of reaction of an enolate ion with an electrophile E^+ are possible. Reaction on carbon to yield an α-substituted carbonyl product is more common.

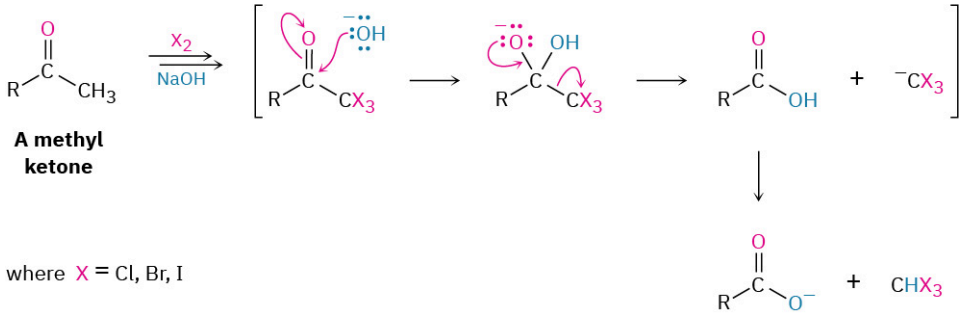
As an example of enolate ion reactivity, aldehydes and ketones undergo base-promoted α halogenation. Even relatively weak bases such as hydroxide ion are effective for halogenation because it's not necessary to convert the ketone completely into its enolate ion. As soon as a small amount of enolate is generated, halogen reacts with it immediately, removing it from the reaction and driving the equilibrium toward further enolate ion

formation.



Base-promoted halogenation of aldehydes and ketones is seldom used in practice because it's difficult to stop the reaction at the monosubstituted product. An α-halogenated ketone is generally more acidic than the starting, unsubstituted ketone because of the electron-withdrawing inductive effect of the halogen atom. Thus, the monohalogenated products are themselves rapidly turned into enolate ions and further halogenated.

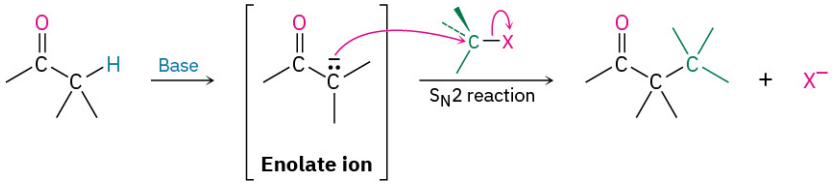
If excess base and halogen are used, a methyl ketone is triply halogenated and then cleaved by base in the *haloform reaction*. The products are a carboxylic acid plus a so-called haloform (chloroform, $CHCl_3$; bromoform, $CHBr_3$; or iodoform, CHI_3). Note that the second step of the reaction is a nucleophilic acyl substitution of $^-CX_3$ by ^-OH. That is, a halogen-stabilized carbanion acts as a leaving group.



PROBLEM 22-9 Why do you suppose ketone halogenations in acidic media are referred to as being acid-*catalyzed*, whereas halogenations in basic media are base-*promoted*? In other words, why is a full equivalent of base required for halogenation?

22.7 Alkylation of Enolate Ions

Perhaps the most useful reaction of enolate ions is their alkylation by treatment with an alkyl halide, thereby forming a new C–C bond and joining two smaller pieces into one larger molecule. Alkylation occurs when the nucleophilic enolate ion reacts with the electrophilic alkyl halide in an S_N2 reaction and displaces the leaving group by backside attack.

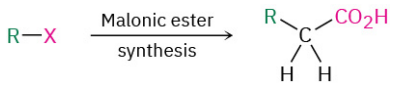


Alkylation reactions are subject to the same constraints that affect all S_N2 reactions (**Section 11.3**). Thus, the leaving group X in the alkylating agent R–X can be chloride, bromide, or iodide. The alkyl group R should be primary or methyl, and preferably should be allylic or benzylic. Secondary halides react poorly, and tertiary halides don't react at all because a competing E2 elimination of HX occurs instead. Vinylic and aryl halides are also unreactive because a backside approach is sterically prevented.

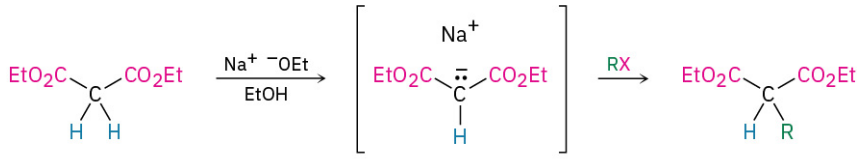
$$R{-}X \begin{cases} -X\text{: Tosylate} > -I > -Br > -Cl \\ R{-}\text{: Allylic} \approx \text{Benzylic} > H_3C{-} > RCH_2{-} \end{cases}$$

The Malonic Ester Synthesis

One of the oldest and best known carbonyl alkylation reactions is the **malonic ester synthesis**, a method for preparing a carboxylic acid from an alkyl halide while lengthening the carbon chain by two atoms.

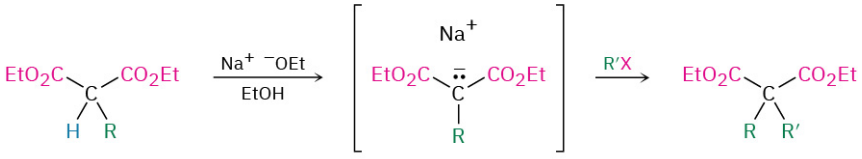


Diethyl propanedioate, commonly called diethyl malonate, or *malonic ester,* is relatively acidic (pK_a = 13) because its α hydrogens are flanked by two carbonyl groups. Thus, malonic ester is easily converted into its enolate ion by reaction with sodium ethoxide in ethanol. The enolate ion, in turn, is a good nucleophile that reacts rapidly with an alkyl halide to give an α-substituted malonic ester. Note in the following examples that the abbreviation "Et" is used for an ethyl group, $-CH_2CH_3$.



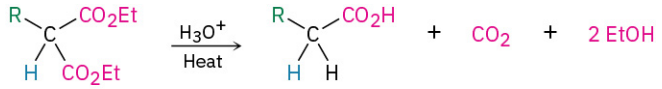
Diethyl propanedioate (malonic ester) **Sodio malonic ester** **An alkylated malonic ester**

The product of a malonic ester alkylation has one acidic α hydrogen remaining, so the alkylation process can be repeated to yield a dialkylated malonic ester.



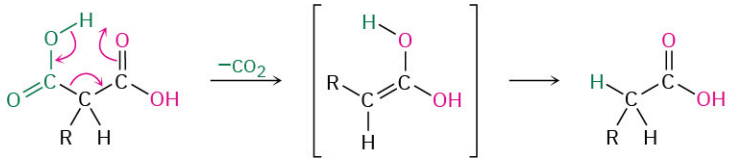
An alkylated malonic ester **A dialkylated malonic ester**

On heating with aqueous hydrochloric acid, the alkylated (or dialkylated) malonic ester undergoes hydrolysis of its two ester groups followed by ***decarboxylation*** (loss of CO_2) to yield a substituted monocarboxylic acid.

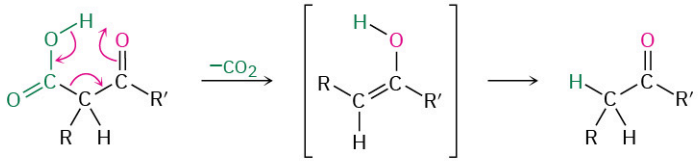


An alkylated malonic ester **A carboxylic acid**

Decarboxylation is not a general reaction of carboxylic acids. Rather, it is unique to compounds that have a second carbonyl group two atoms away from the $-CO_2H$. That is, only substituted malonic acids and β-keto acids undergo loss of CO_2 on heating. The decarboxylation reaction occurs by a cyclic mechanism and involves initial formation of an enol, thereby accounting for the need to have a second carbonyl group appropriately positioned.

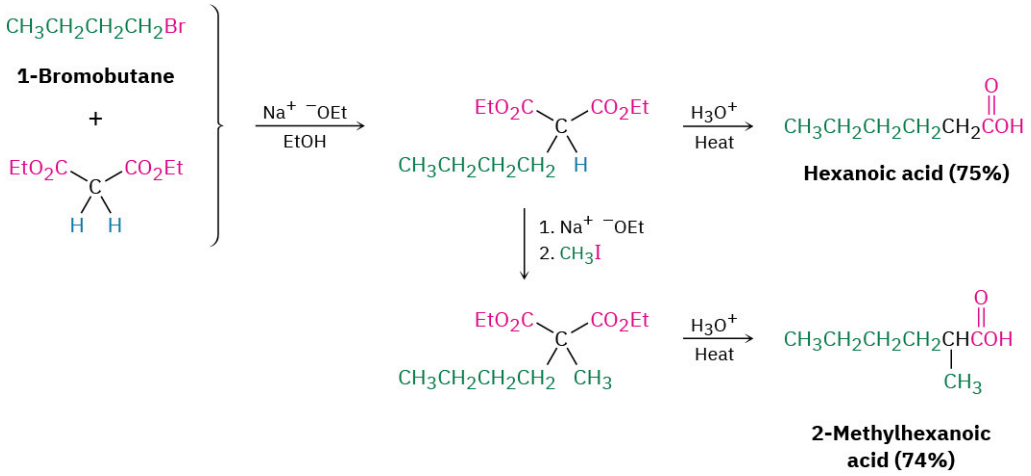


A diacid **An acid enol** **A carboxylic acid**

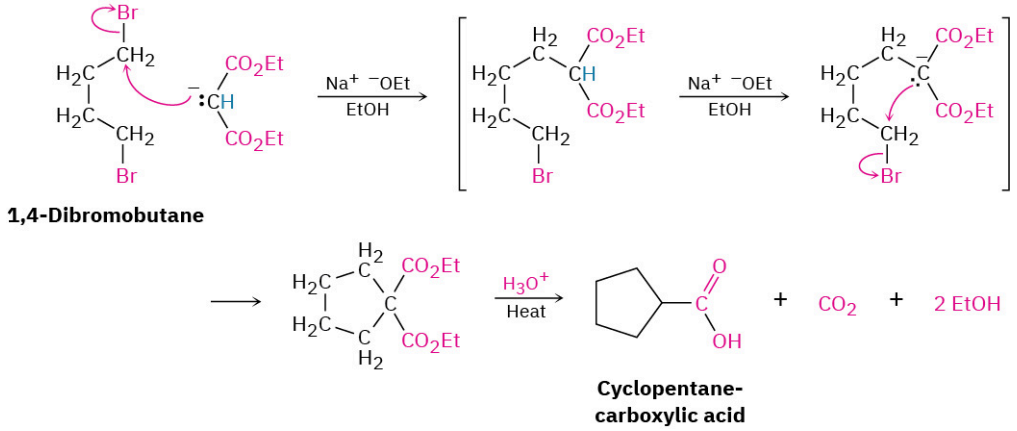


A β-keto acid **An enol** **A ketone**

As noted previously, the overall effect of malonic ester synthesis is to convert an alkyl halide into a carboxylic acid while lengthening the carbon chain by two atoms ($RX \rightarrow RCH_2CO_2H$).



A malonic ester synthesis can also be used to prepare *cyclo*alkanecarboxylic acids. For example, when 1,4-dibromobutane is treated with diethyl malonate in the presence of two equivalents of sodium ethoxide base, the second alkylation step occurs intramolecularly to yield a cyclic product. Hydrolysis and decarboxylation then give cyclopentanecarboxylic acid. Three-, four-, five-, and six-membered rings can be prepared in this way, but yields decrease for larger ring sizes.



WORKED EXAMPLE 22.2

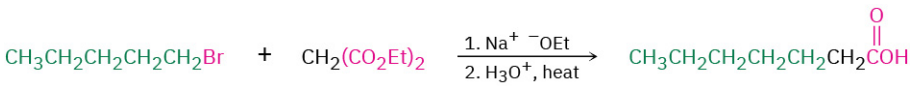
Using Malonic Ester Synthesis to Prepare a Carboxylic Acid

How would you prepare heptanoic acid using a malonic ester synthesis?

Strategy

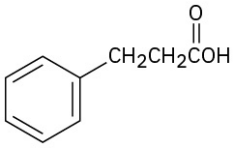
A malonic ester synthesis converts an alkyl halide into a carboxylic acid having two more carbons. Thus, a seven-carbon acid chain must be derived from the five-carbon alkyl halide 1-bromopentane.

Solution

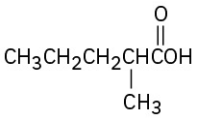


PROBLEM 22-10 How could you use a malonic ester synthesis to prepare the following compounds? Show all steps.

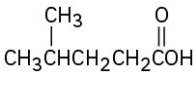
(a)



(b)

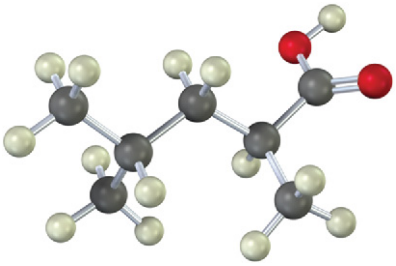


(c)



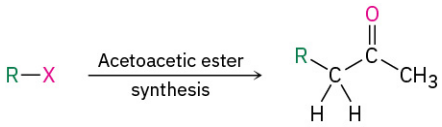
PROBLEM 22-11 Monoalkylated and dialkylated acetic acids ($R_3C–CO_2H$) can't be prepared by malonic ester synthesis, but trialkylated acetic acids ($R_3C–CO_2H$) can't be prepared. Explain.

PROBLEM 22-12 How could you use a malonic ester synthesis to prepare the following compound?

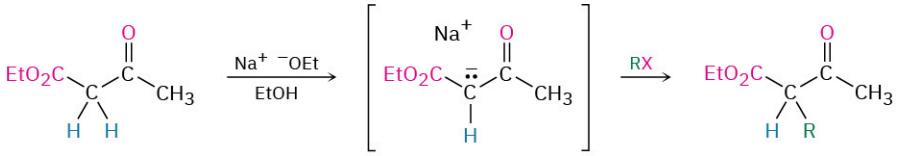


The Acetoacetic Ester Synthesis

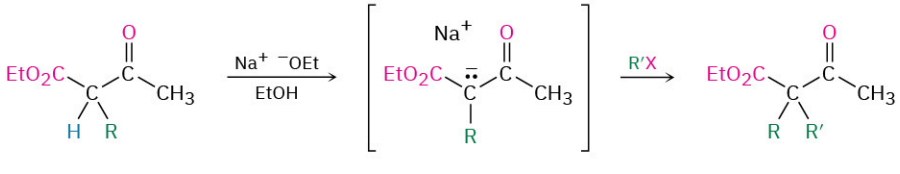
Just as the malonic ester synthesis converts an alkyl halide into a carboxylic acid, the **acetoacetic ester synthesis** converts an alkyl halide into a methyl ketone having three more carbons.



Ethyl 3-oxobutanoate, commonly called ethyl acetoacetate, or *acetoacetic ester,* is much like malonic ester in that its α hydrogens are flanked by two carbonyl groups. It is therefore readily converted into its enolate ion, which can be alkylated by reaction with an alkyl halide. A second alkylation can also be carried out if desired, since acetoacetic ester has two acidic α hydrogens.

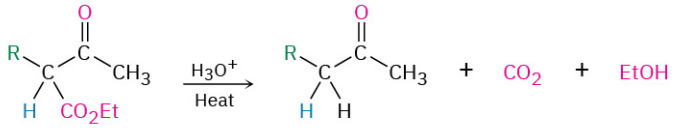


Ethyl acetoacetate (acetoacetic ester) | **Sodio acetoacetic ester** | **A monoalkylated acetoacetic ester**



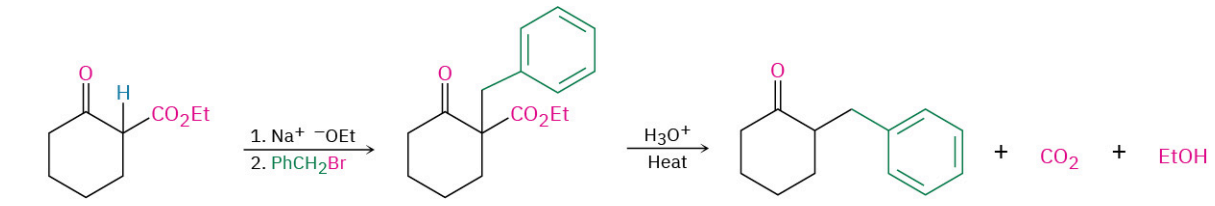
A monoalkylated acetoacetic ester | **A dialkylated acetoacetic ester**

On heating with aqueous HCl, the alkylated (or dialkylated) acetoacetic ester is hydrolyzed to a β-keto acid, which then undergoes decarboxylation to yield a ketone product. The decarboxylation occurs in the same way as in the malonic ester synthesis and involves a ketone enol as the initial product.



An alkylated acetoacetic ester | **A methyl ketone**

The three-step sequence of (**1**) enolate ion formation, (**2**) alkylation, and (**3**) hydrolysis/decarboxylation is applicable to all β-keto esters with acidic α hydrogens, not just to acetoacetic ester itself. For example, cyclic β-keto esters, such as ethyl 2-oxocyclohexanecarboxylate, can be alkylated and decarboxylated to give 2-substituted cyclohexanones.



Ethyl 2-oxocyclohexane-carboxylate (a cyclic β-keto ester)

2-Benzylcyclohexanone (77%)

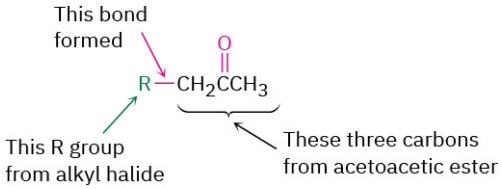
WORKED EXAMPLE 22.3

Using Acetoacetic Ester Synthesis to Prepare a Ketone

How would you prepare 2-pentanone by an acetoacetic ester synthesis?

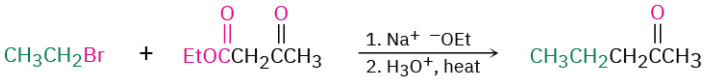
Strategy

An acetoacetic ester synthesis yields a methyl ketone by adding three carbons to an alkyl halide.



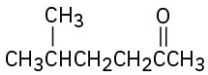
Thus, the acetoacetic ester synthesis of 2-pentanone must involve reaction of bromoethane.

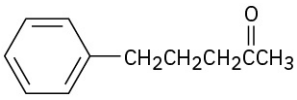
Solution



2-Pentanone

PROBLEM 22-13 What alkyl halides would you use to prepare the following ketones by an acetoacetic ester synthesis?

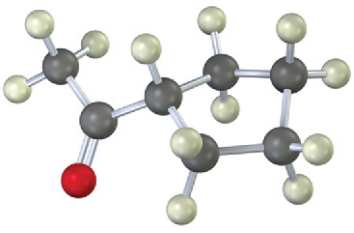
(a) 

(b) 

PROBLEM 22-14 Which of the following compounds can't be prepared by an acetoacetic ester synthesis? Explain.

(a) Phenylacetone **(b)** Acetophenone **(c)** 3,3-Dimethyl-2-butanone

PROBLEM 22-15 How would you prepare the following compound using an acetoacetic ester synthesis?



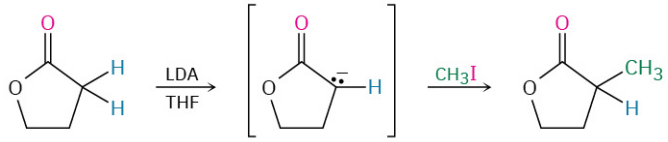
Direct Alkylation of Ketones, Esters, and Nitriles

Both the malonic ester synthesis and the acetoacetic ester synthesis are easy to carry out because they involve relatively acidic dicarbonyl compounds. As a result, sodium ethoxide in ethanol can be used to prepare the

necessary enolate ions. Alternatively, however, it's also possible in many cases to directly alkylate the α position of *mono*carbonyl compounds. A strong, sterically hindered base such as LDA is needed so that complete conversion to the enolate ion takes place rather than a nucleophilic addition, and a nonprotic solvent must be used.

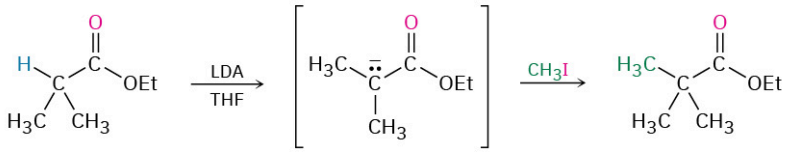
Ketones, esters, and nitriles can all be alkylated using LDA or related dialkylamide bases in tetrahydrofuran (THF). Aldehydes, however, rarely give high yields of pure products because their enolate ions undergo carbonyl condensation reactions instead of alkylation. (We'll study this condensation reaction in the next chapter.) Some specific examples of alkylation reactions are shown.

Lactone



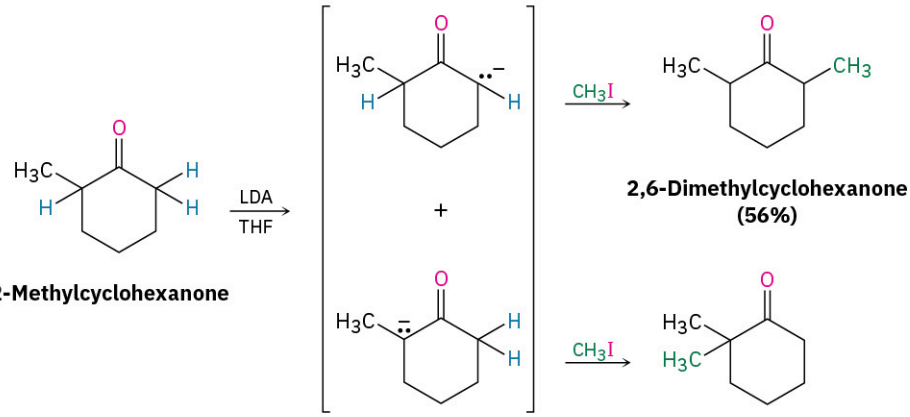
Butyrolactone **2-Methylbutyrolactone (88%)**

Ester



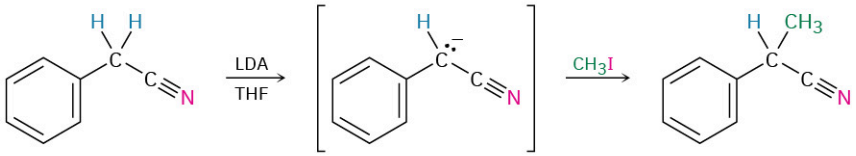
Ethyl 2-methylpropanoate **Ethyl 2,2-dimethylpropanoate (87%)**

Ketone



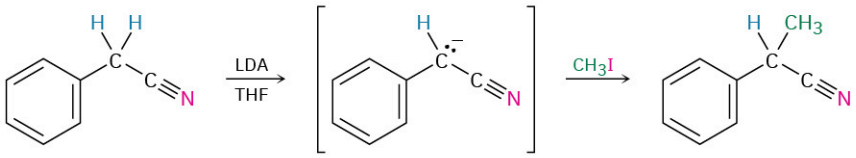
2-Methylcyclohexanone **2,6-Dimethylcyclohexanone (56%)** **2,2-Dimethylcyclohexanone (6%)**

Nitrile



Phenylacetonitrile **2-Phenylpropane-nitrile (71%)**

Nitrile



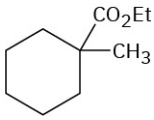
Phenylacetonitrile **2-Phenylpropane-nitrile (71%)**

Note in the ketone example that alkylation of 2-methylcyclohexanone leads to a mixture of products because both possible enolate ions are formed. In general, the major product in such cases occurs by alkylation at the less hindered, more accessible position. Thus, alkylation of 2-methylcyclohexanone occurs primarily at C6 (secondary) rather than C2 (tertiary).

WORKED EXAMPLE 22.4

Using an Alkylation Reaction to Prepare a Substituted Ester

How might you use an alkylation reaction to prepare ethyl 1-methylcyclohexanecarboxylate?

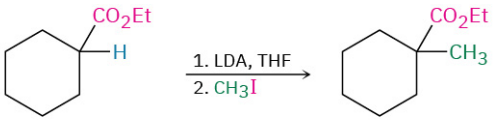


Ethyl 1-methylcyclohexanecarboxylate

Strategy

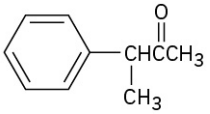
An alkylation reaction is used to introduce a methyl or primary alkyl group onto the α position of a ketone, ester, or nitrile by S_N2 reaction of an enolate ion with an alkyl halide. Thus, we need to look at the target molecule and identify any methyl or primary alkyl groups attached to an α carbon. In the present instance, the target has an α methyl group, which might be introduced by alkylation of an ester enolate ion with iodomethane.

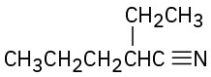
Solution

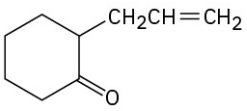


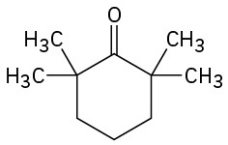
Ethyl cyclohexane-carboxylate **Ethyl 1-methylcyclo-hexanecarboxylate**

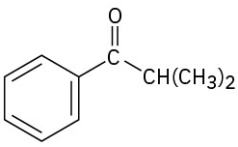
PROBLEM 22-16 Show how you might prepare the following compounds using an alkylation reaction as the key step:

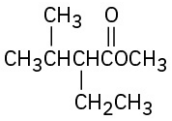
(a) 

(b) 

(c) 

(d) 

(e) 

(f) 

Biological Alkylations

Alkylations are rare but not unknown in biological chemistry. One example occurs during biosynthesis of the antibiotic indolmycin from indolylpyruvate when a base abstracts an acidic hydrogen from an α position and the resultant enolate ion carries out an S_N2 alkylation reaction on the methyl group of *S*-adenosylmethionine (SAM; **Section 11.6**). Although it's convenient to speak of "enolate ion" intermediates in biological pathways, it's unlikely that they exist for long in an aqueous cellular environment. Rather, proton removal and alkylation

probably occur at essentially the same time (FIGURE 22.7).

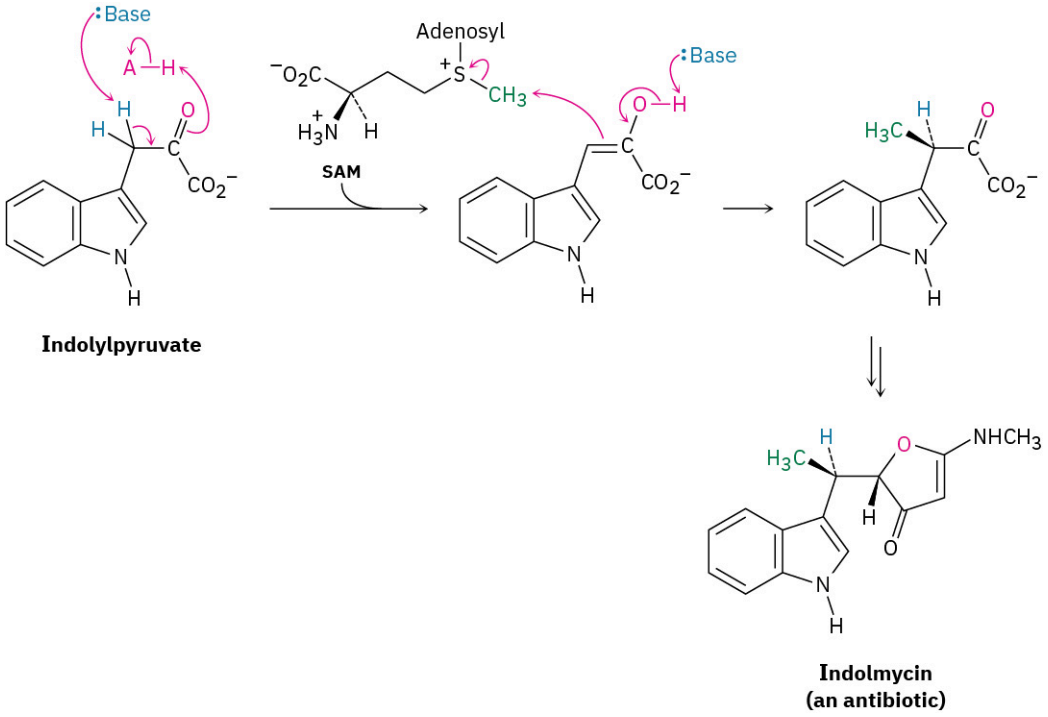


FIGURE 22.7 **The biosynthesis of indolmycin from indolylpyruvate** occurs through a pathway that includes an alkylation reaction of a short-lived enolate ion intermediate.

CHEMISTRY MATTERS

Barbiturates

The use of herbal remedies to treat illness and disease goes back thousands of years, but the medical use of chemicals prepared in the laboratory has a much shorter history. Barbiturates, a large class of drugs with a wide variety of uses, constitute one of the earliest successes of medicinal chemistry. The synthesis and medical use of barbiturates goes back to 1904 when Bayer, a German chemical company, first marketed a compound called barbital, trade named Veronal, as a treatment for insomnia. Since that time, more than 2500 different barbiturate analogs have been synthesized by drug companies, more than 50 have been used medicinally, and about a dozen are still in use as anesthetics, anticonvulsants, sedatives, and anxiolytics.



FIGURE 22.8 Different barbiturates come in a multitude of colors, giving rise to similarly colorful street names when the drugs are abused. (credit: "Medications" by freestocks.org/Flickr, Public Domain)

The synthesis of barbiturates is relatively simple and relies on reactions that are now familiar: enolate alkylations and nucleophilic acyl substitutions. Starting with diethyl malonate, alkylation of the corresponding enolate ion with simple alkyl halides provides a wealth of different disubstituted malonic esters. Reaction with urea, $(H_2N)_2C{=}O$, then gives the product barbiturates by a twofold nucleophilic acyl substitution reaction of the ester groups with the $-NH_2$ groups of urea (**FIGURE 22.9**). Amobarbital (Amytal), pentobarbital (Nembutal), and secobarbital (Seconal) are typical examples.

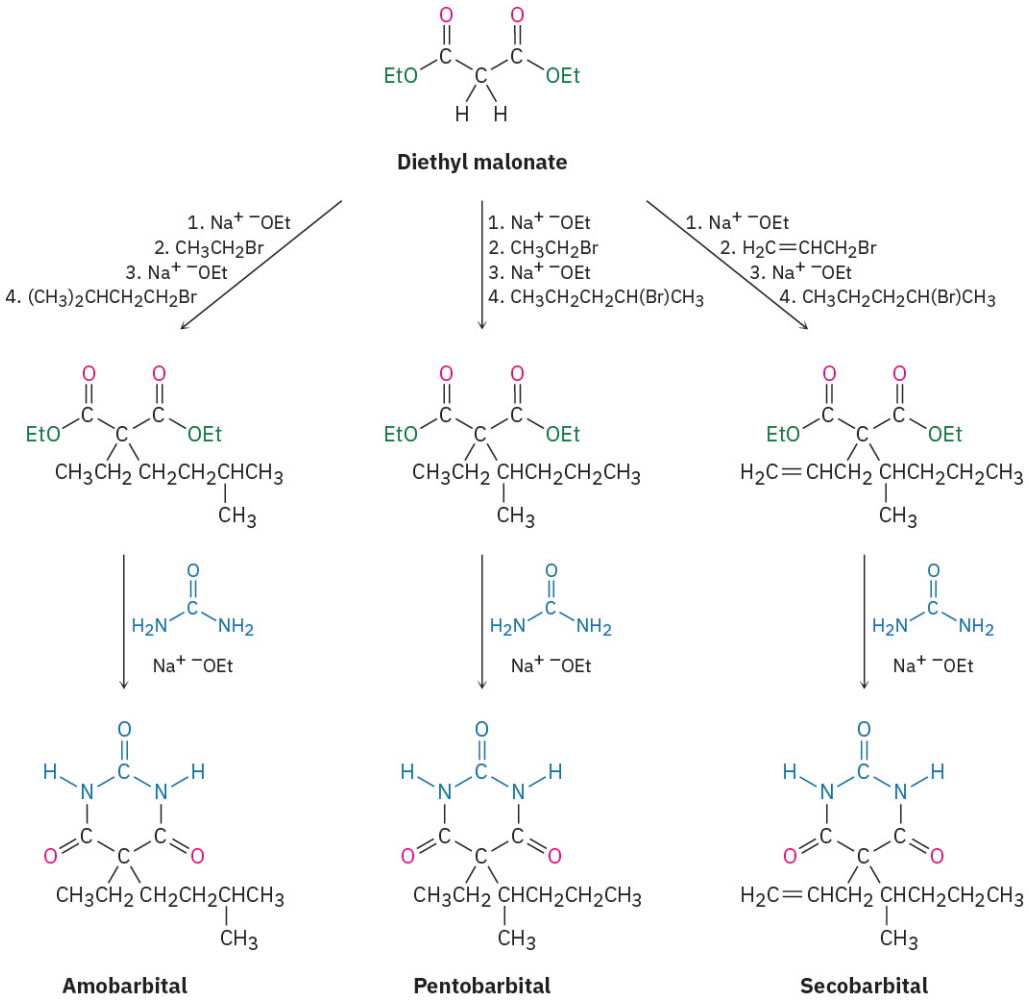
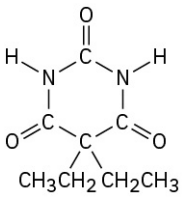


FIGURE 22.9 The synthesis of barbiturates relies on malonic ester alkylations and nucleophilic acyl substitution reactions. More than 2500 different barbiturates have been synthesized over the past 100 years.



Barbital (Veronal), the first barbiturate

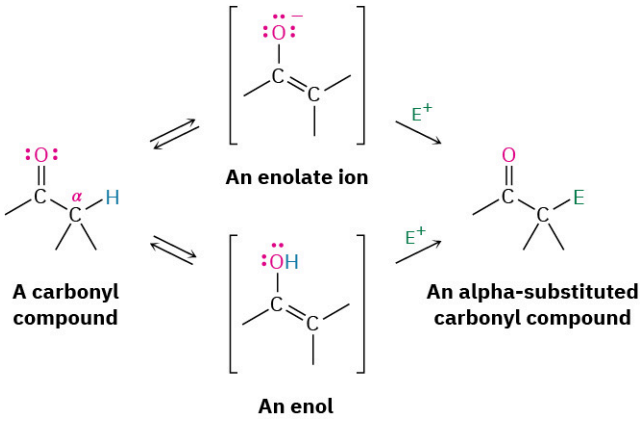
In addition to their prescribed medical uses, many barbiturates have also found widespread illegal use as street drugs. Each barbiturate often comes as a tablet of regulated size, shape, and color. Although still used today, most barbiturates have been replaced by safer, more potent alternatives with markedly different structures.

Key Terms

- **acetoacetic ester synthesis**
- **α-substitution reaction**
- **decarboxylation**
- **enol**
- **enolate ion**
- **malonic ester synthesis**
- **tautomer**

Summary

The α-substitution reaction of a carbonyl compound through either an **enol** or **enolate ion** intermediate is one of the four fundamental reaction types in carbonyl-group chemistry.

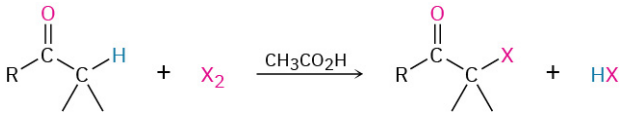


Carbonyl compounds are in an equilibrium with their enols, a process called keto–enol tautomerism. Although enol **tautomers** are normally present to only a small extent at equilibrium and can't usually be isolated in pure form, they nevertheless contain a highly nucleophilic double bond and react with electrophiles in an ***α*-substitution reaction**. An example is the *α* halogenation of ketones on treatment with Cl_2, Br_2, or I_2 in acid solution. Alpha bromination of carboxylic acids can be similarly accomplished by the Hell–Volhard–Zelinskii (HVZ) reaction, in which an acid is treated with Br_2 and PBr_3. The *α*-halogenated products can then undergo base-induced E2 elimination to yield *α*,*β*-unsaturated carbonyl compounds.

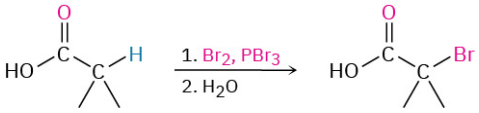
Alpha hydrogen atoms of carbonyl compounds are weakly acidic and can be removed by strong bases, such as lithium diisopropylamide (LDA), to yield nucleophilic enolate ions. The most useful reaction of enolate ions is their S_N2 alkylation with alkyl halides. The **Malonic ester synthesis** converts an alkyl halide into a carboxylic acid with the addition of two carbon atoms ($RX \rightarrow RCH_2CO_2H$). Similarly, the **acetoacetic ester synthesis** converts an alkyl halide into a methyl ketone with the addition of three carbon atoms ($RX \rightarrow RCH_2COCH_3$). In addition, many carbonyl compounds, including ketones, esters, and nitriles, can be directly alkylated by treatment with LDA and an alkyl halide.

Summary of Reactions

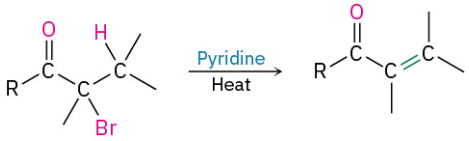
1. Aldehyde/ketone halogenation (Section 22.3)



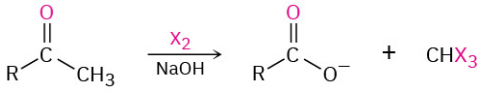
2. Hell–Volhard–Zelinskii bromination of acids (Section 22.4)



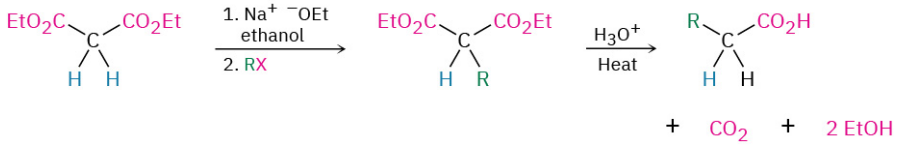
3. Dehydrobromination of *α*-bromo ketones (Section 22.3)



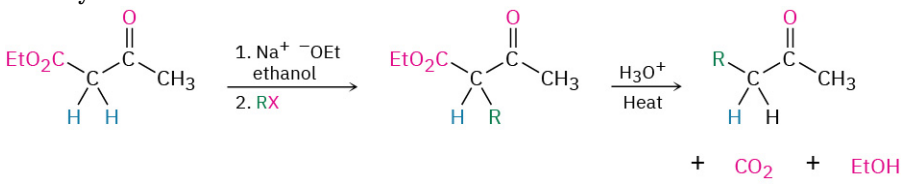
4. Haloform reaction (Section 22.6)



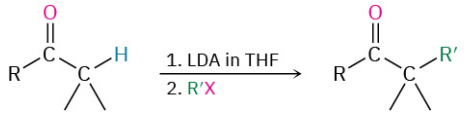
5. Alkylation of enolate ions (Section 22.7)
   a. Malonic ester synthesis



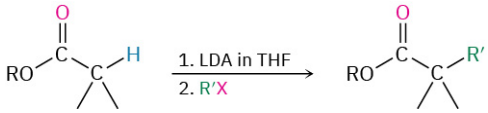
b. Acetoacetic ester synthesis



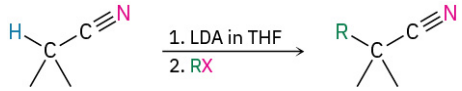
c. Direct alkylation of ketones



d. Direct alkylation of esters



e. Direct alkylation of nitriles

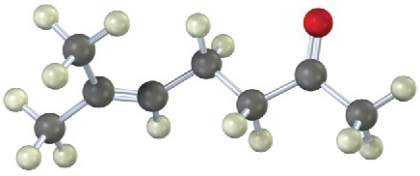


Additional Problems

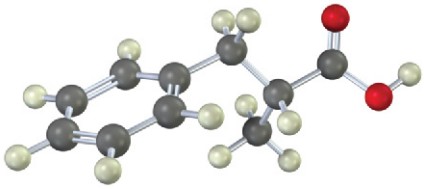
Visualizing Chemistry

PROBLEM 22-17 Show the steps in preparing each of the following substances using either a malonic ester synthesis or an acetoacetic ester synthesis:

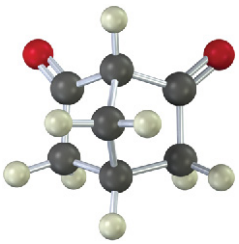
(a)



(b)



PROBLEM 22-18 Unlike most β-diketones, the following β-diketone has no detectable enol content and is about as acidic as acetone. Explain.



PROBLEM 22-19 For a given α hydrogen atom to be acidic, the C–H bond must be parallel to the p orbitals of the C=O double bond (that is, perpendicular to the plane of the adjacent carbonyl group). Identify the most acidic hydrogen atom in the conformation shown for the following structure. Is it axial or equatorial?